Seminar 03 cheat sheet

Gradient Descent and Automatic Differentiation

Optimization Methods
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Jensen's inequality

- **Convex** f on a convex set: $f\big((1-t)x+ty\big) \le (1-t)f(x)+tf(y)$ for all x, y and $t \in [0,1]$; the chord lies on or above the graph, no derivatives needed. **Strictly convex**: strict for $x \ne y$, $t \in (0,1)$. For x^2 the gap is $t(1-t)(x-y)^2$.
- **Three equivalent tests** on an open convex set: chord, the definition; tangent $f(y) \ge f(x) + \nabla f(x)^\top (y-x)$ for a differentiable f; curvature $\nabla^2 f(x) \succeq 0$ for a twice differentiable f. Strict convexity follows from $\nabla^2 f \succ 0$ everywhere, not conversely: $|x|^3$.
- **Weights** $t_i \ge 0$, $\sum_i t_i = 1$: $f\big(\sum_i t_i x_i\big) \le \sum_i t_i f(x_i)$; also $f(\mathbb{E}X) \le \mathbb{E} f(X)$; reversed for concave functions: $\log \mathbb{E}X \ge \mathbb{E} \log X$. **One pair disproves** convexity: x^3 at $x=-2$, $y=1$, $t=0.5$ has the graph (-0.125) above the chord (-3.5).

Constants μ, L and κ

- μ**-strongly convex**: $f - \frac{\mu}{2}\|\cdot\|^2$ is convex, or $\nabla^2 f \succeq \mu I$. L**-smooth**: $\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\|$, or $-LI \preceq \nabla^2 f \preceq LI$.
- **Two parabolas**: $\frac{\mu}{2}\|y-x\|^2 \le f(y) - f(x) - \nabla f(x)^\top (y-x) \le \frac{L}{2}\|y-x\|^2$. In Jensen form the gap between chord and graph lies between $\frac{\mu}{2}t(1-t)\|x-y\|^2$ and $\frac{L}{2}t(1-t)\|x-y\|^2$.
- **Quadratic** $f = \frac{1}{2}x^\top Q x - b^\top x$, $Q \succ 0$: $\nabla f = Qx - b$, $\mu = \lambda_{\min}(Q)$, $L = \lambda_{\max}(Q)$. **Condition number** $\kappa = L/\mu \ge 1$; a large κ is a long narrow valley.
- **PL condition** (Polyak–Łojasiewicz): $\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu\big(f(x) - f^*\big)$ for all x. It follows from μ-strong convexity and makes every stationary point a global minimizer; convexity is not needed $(x^2 + 3\sin^2 x)$ nor sufficient (x^4).

Gradient descent

- **Iteration** $x_{k+1} = x_k - \alpha \nabla f(x_k)$ along the steepest descent direction $-\nabla f(x_k)$; d is a descent direction if $\nabla f(x)^\top d < 0$.
- **Descent lemma** (L-smooth f): $f(y) \le f(x) + \nabla f(x)^\top (y-x) + \frac{L}{2}\|y-x\|^2$. One step: $f(x^+) \le f(x) - \alpha\big(1 - \frac{L\alpha}{2}\big)\|\nabla f(x)\|^2$, a decrease for $0 < \alpha < 2/L$; the **safe step** $1/L$ gives $f(x^+) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$.
- **Quadratic**, eigen-coordinates: $z_{k,i} = (1-\alpha\lambda_i)^k z_{0,i}$. Convergence for every $x_0 \iff 0 < \alpha < 2/L$. The **stability boundary** $2/L$ is excluded: there the stiff coordinate flips sign forever.
- **Contraction factor** $\rho(\alpha) = \max\big(|1-\alpha\mu|, |1-\alpha L|\big)$: $\|x_k - x^*\| \le \rho^k \|x_0 - x^*\|$, $f(x_k) - f^* \le \rho^{2k}\big(f(x_0) - f^*\big)$, $\rho(1/L) = 1 - 1/\kappa$. **Optimal step** $\frac{2}{L+\mu}$: $\rho = \frac{\kappa-1}{\kappa+1}$.

Rates

- **Convex**, $\alpha = 1/L$: $f(x_k) - f^* \le \frac{L\|x_0 - x^*\|^2}{2k} = O(1/k)$, since the key inequality $f(x_{k+1}) - f^* \le \frac{L}{2}\big(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2\big)$ telescopes.
- μ**-strongly convex** or **PL**, $\alpha = 1/L$: $f(x_k) - f^* \le (1-\mu/L)^k \big(f(x_0) - f^*\big)$, from $f(x_{k+1}) \le f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$ and PL; convexity is not used.
- **Iterations** to cut the gap by the factor ε: $k \ge \frac{\ln \varepsilon}{\ln(1-1/\kappa)}$, about $\kappa \ln(1/\varepsilon)$. For $\kappa = 1000$, $\varepsilon = 10^{-6}$: 13809 at $1/L$, 3454 at $\frac{2}{L+\mu}$ on a quadratic. A bound, not a forecast: runs are often faster.

Armijo backtracking

- **Armijo condition**: $f(x + \alpha d) \le f(x) + c\alpha \nabla f(x)^\top d$; for $d = -\nabla f$ the right-hand side is $f(x) - c\alpha\|\nabla f(x)\|^2$. **Backtracking**: $\alpha = \alpha_0, \beta\alpha_0, \beta^2\alpha_0, \dots$ until it holds; typically $\beta = \frac{1}{2}$, $c = 10^{-4}$.
- **Termination** (L-smooth f): the condition holds for all $\alpha \le 2(1-c)/L$, so the loop stops with $\alpha \ge \min\big(\alpha_0, 2\beta(1-c)/L\big)$. Check $\nabla f(x)^\top d < 0$ and cap the reductions.

Stopping rules

- **Scaled tests**: $\|\nabla f(x_k)\| \le \varepsilon_g \max\big(1, |f(x_k)|\big)$, $\|x_{k+1} - x_k\| \le \varepsilon_x \max\big(1, \|x_k\|\big)$, $|f(x_{k+1}) - f(x_k)| \le \varepsilon_f \max\big(1, |f(x_k)|\big)$; in code $\text{atol} + \text{rtol} \cdot \max(1, s)$.
- The change tests shrink with the step, $\|x_{k+1} - x_k\| = \alpha_k \|\nabla f(x_k)\|$, so a tiny step fires them anywhere. **Combined rule**: success only on the gradient test; a change test alone signals stagnation; cap the iterations and stop when f or ∇f is not finite.

Automatic differentiation

- **Computational graph**: intermediate variables in topological order; $z = \sin(xy) + x^2$: $u = xy$, $v = \sin u$, $w = x^2$, $z = v + w$. The **forward pass** computes and stores the values.
- **Reverse pass**: the adjoint $\bar{v} = \partial z / \partial v$; $\bar{z} = 1$, the others 0; visit the nodes in reverse topological order and for each edge $u \to v$ do $\bar{u}$ += $\bar{v}\,\partial v / \partial u$. Here $\bar{x} = y\cos u + 2x$, $\bar{y} = x \cos u$.
- **Accumulation**: a node feeding several others collects one term per edge, $\bar{v}_i = \sum_{j:\, i \to j} \bar{v}_j \, \partial v_j / \partial v_i$; $z = x^2 + x^3$ gives $\bar{x} = 2x + 3x^2$. Assignment keeps only the last term.
- **Forward mode** gives the Jacobian–vector product $J\dot{x}$ (JVP): n sweeps for a gradient in $\mathbb{R}^n$. **Reverse mode** gives the vector–Jacobian product $\bar{z}^\top J$ (VJP): the whole gradient of a scalar in one sweep. A sweep costs a small multiple of one evaluation of f; reverse mode stores every intermediate value (the tape).

Finite differences

- **Central difference** $\frac{f(x+he_i) - f(x-he_i)}{2h}$, error $\approx \frac{h^2}{6}|\partial_i^3 f| + \frac{\varepsilon |f|}{h}$: truncation plus round-off, ε the machine epsilon. Best $h \sim \varepsilon^{1/3} \approx 6 \cdot 10^{-6}$, error $\sim \varepsilon^{2/3}$. A check of a gradient, not a way to compute it.

Logistic loss

- **Derivative**: $\frac{d\ell}{ds} = \sigma(s) - y$ for $\ell = \log(1 + e^s) - ys$, $s = w^\top x + b$, $y \in \{0, 1\}$, $\sigma(s) = \frac{1}{1+e^{-s}}$; so $\nabla_w \ell = \big(\sigma(s) - y\big)x$ and $\frac{\partial \ell}{\partial b} = \sigma(s) - y$.
- **Smoothness** of the mean loss over the rows of X: $\nabla^2 f = \frac{1}{n} X^\top \operatorname{diag}\big(\sigma_i(1-\sigma_i)\big) X$ and $\sigma(1-\sigma) \le \frac{1}{4}$, so f is convex with $L \le \frac{\lambda_{\max}(X^\top X)}{4n}$, attained at $w = 0$.

Top traps

- At $\alpha = 2/L$ convergence already fails: it is a boundary, not a step to use; the limit comes from $L = \lambda_{\max}$, and μ only sets the speed.
- Armijo with $+c\alpha\|\nabla f\|^2$ accepts steps that increase f; a small change in x or f can mean a tiny step, not a solution.
- Reverse pass: = instead of += loses terms; $\sigma(s) - y$ is $d\ell/ds$, and the gradient in w carries the factor x.

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